

• arXiv: 2504.02882 • Tool-Augmented LLMs

DiaTool-DPO: Multi-Turn Direct Preference Optimization for Tool-Augmented LLMs

Modeling TA-LLM interactions as a 5-state MDP and automatically constructing preference trajectory pairs to improve slot-filling and tool call rejection — without any human annotation.

Direct Preference Optimization

Markov Decision Process

FunctionChat-Bench

Near GPT-4o Performance

PROBLEM • MOTIVATION

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Modeling TA-LLM Dialogue as a 5-State MDP

S ₁	Initial State No conversation history. Out-of-scope requests return a rejection and remain here (Type 3 self-loop).
S ₂	Tool Selected — Incomplete Slots Tool is identified but required parameters are missing. Slot-filling QA interactions occur here; may repeat N times.
S ₃	Tool Selected — All Slots Complete Both tool selection and all required argument values are fully determined. Ready for execution.
S ₄	Wait for Tool Response Tool call has been dispatched. System awaits results from the external tool API.
S ₅	Complete Tool results received. Final completion message generated and returned to the user.

QUERY CLASSIFICATION

Three Query Types Define Distinct State Trajectories

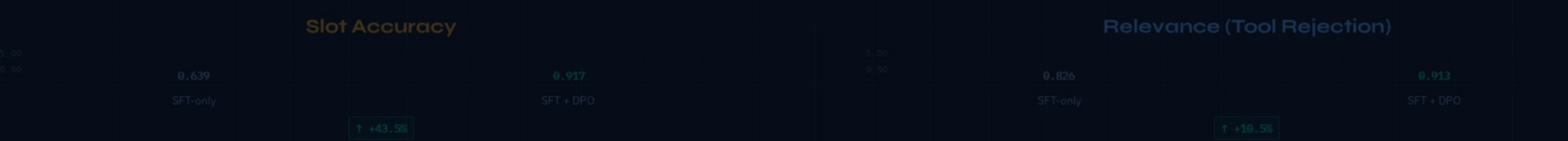
Query Type	State Trajectory	Example
Query 1	Successful query	SELECT * FROM table1
Query 2	All required arguments present	SELECT * FROM table1 WHERE column1 = 'value1'
Query 3	One or more required arguments missing	SELECT * FROM table1 WHERE column1 = 'value1' AND column2 = 'value2'
Query 4	No matching tool available	SELECT * FROM table1 WHERE column1 = 'value1' AND column2 = 'value2' AND column3 = 'value3'

Automatic Paired Trajectory Construction – No Human Labels

QUERY	✓ CHOSEN TRAJECTORY	✗ REJECTED TRAJECTORY	LEARNING LESSON
Type 1	$S_1 \rightarrow S_3 \rightarrow S_4 \rightarrow S_5$	$S_1 \rightarrow S_2 \rightarrow S_3 \rightarrow S_4 \rightarrow S_5$ $S_2 \rightarrow (S_2 \times N) \rightarrow S_3 \rightarrow S_4 \rightarrow S_5$	Prevent redundant slot-filling questions when all required arguments are already present
Type 2	$S_1 \rightarrow (S_2 \times N) \rightarrow S_3 \rightarrow S_4 \rightarrow S_5$	$S_1 \rightarrow S_2 \rightarrow S_4 \rightarrow S_5$ $S_2 \rightarrow (S_2 \times M) \rightarrow S_3 \rightarrow S_4 \rightarrow S_5 \ (M \neq N)$	Prevent slot hallucination; gather exactly the missing arguments — no more, no less
Type 3	$S_1 \rightarrow S_1$ <i>(graceful rejection)</i>	$S_1 \rightarrow S_2 \rightarrow S_4$ $S_1 \rightarrow (S_2 \times N) \rightarrow S_3 \rightarrow S_4$	Reject out-of-scope requests; never hallucinate a tool call for unavailable functionality

+43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO

SFT-only vs. SFT + DiaTool-DPO on FunctionChat-Bench. LLaMA-3-8B shows the most dramatic improvement.



KEY TAKEAWAYS

RL-Based Dialogue Control for Production TA-LLMs

- 01

MDP formulation enables systematic preference pair construction. Defining 5 latent states and 3 query types allows fully automatic generation of chosen/rejected trajectory pairs — no human annotation required.
- 02

DPO complements SFT — it does not replace it. DiaTool-DPO-only performs poorly across all models. Gains come from SFT + DPO together, confirming DPO is a fine-grained corrective signal, not a standalone training recipe.

03

Slot-filling and tool rejection see the largest gains. DiaTool-DPO improves slot accuracy by 1.5% (0.019 → 0.021) and rejection by 10.5% (0.024 → 0.034). The largest gains belong to SFT + DPO.